

Archita Sukhwani

📍 Bellevue, WA | 📞 +1 (858) 241-4810, +91 7083586081 | ✉️ sukhwaniarchita@gmail.com | [in linkedin.com/architasukhwani](https://www.linkedin.com/in/architasukhwani)

Technical Skills

Languages: Python, Scala, C, C++, MySQL, Shell Scripting

Machine Learning Frameworks: TensorFlow, PyTorch, Keras, Scikit-Learn, NLTK, SpaCy, Hugging Face, Pandas, NumPy

Machine Learning Techniques: Deep Learning, Generative AI, Natural Language Processing, Large Language Models (LLMs)

Database and Container Management: Redis, BigQuery, Docker, Kubernetes

Version Control and Workflow Management: Git, Apache Airflow, Databricks

Work Experience

Oracle India Pvt Ltd

Senior Applied Scientist

Aug 2025 - Present

Hyderabad, India

- Deployed Referrals and Revisits **order-capture** feature to production within the Orders Creation Agent, scaled to **20 customers and 263 physicians**, reducing manual workload and saving physician time via automated capture.
- Designed and developed an Advisor Agent API using **Azure OpenAI**, providing **verbalization recommendations** to physicians to enhance the performance of the Orders Creation Agent

Couture.ai

Data Scientist 3 (Apr 2024 - Aug 2025)

Jul 2020 - Aug 2025

Bengaluru, India

- Lead the development of an advanced **GIST framework** based embedding model to enhance feature representation and vector search accuracy, with an estimated **2.3% revenue lift**.
- Trained a large **sentence transformer** model using contrastive MNR loss with **DeepSpeed** for efficient multi-GPU training, achieving test set **precision of 90+%**.

Data Scientist 2 (Aug 2022 - Apr 2024)

Bengaluru, India

- Reduced **empty search results by 3.5%** and increased **Click-Through Rate (CTR) by 1.8%**, enriching user experience.
- Achieved sub-30ms response times by optimizing the subsequence matching algorithm with sparse matrix computations and customized **FastText embeddings** for **context-based auto-complete**.
- Designed a **context-aware spell correction system** combining SymSpell and fine-tuned **BERT** with Masked Language Modeling (MLM) and Next Sentence Prediction (NSP), improving **accuracy to 91%**.
- Interviewed 25+ candidates** and **mentored 4+ new hires** on semantic search and Scala-Spark performance optimizations.

Associate Data Scientist (Jun 2021 - Jul 2022)

Bengaluru, India

- Achieved a **4% revenue boost** in test vs control groups through enhancements to e-commerce semantic search, elevating the **user intent and query semantics understanding** for vernacular queries.
- Engineered real-time query processing logic powered by an offline corpus, enhancing query understanding and relevance for **over 5.5M products and 50M-70M users of Reliance Retail**.
- Increased **entity detection accuracy by 7%** by incorporating category-level probability distributions into **token entity corpus**.
- Implemented **K-Means Clustering** to **identify brand alternatives** across platforms, driving a **1% revenue increase**.

Data Science Intern (Jul 2020 - Jun 2021)

Bengaluru, India

- Designed a **phonetic- and distance-based spell correction system** using user spell-error probability distribution to achieve **85+% accuracy** on user queries.
- Conducted **keyword mining** from product names using Parts Of Speech (POS) tag patterns to upgrade phrase detection in user queries.

Bhaskaracharya Institute of Space Applications and Geo-Informatics

Data Science Intern

May 2018 - Jul 2018

Gandhinagar, India

- Enhanced newspaper article digitization using OCR and **binarization and noise-reduction** preprocessing techniques via **PyTesseract**, boosting accuracy and efficiency.

Projects

LLM Prompt Recovery

Apr 2024

- Created a system to identify the original prompt that would generate a given text response from a language model.
- Refined Mistral 7B using PEFT and LoRA, and employed mean prompt customization during inference to heighten the accuracy and relevance.

Hallucination Detection in LLMs

Mar 2024

- Developed a hallucination classifier to evaluate responses generated by Mistral 7B in response to instructions from the Open-Orca dataset.
- Applied TF-IDF Vectorizer and an XGBoost Classifier to accurately identify and label hallucinated content.

Education

BITS Pilani

Master of Science in Mathematics

Bachelor of Engineering in Computer Science

Jun 2021

Goa, India

Achievements

- Ranked in **top 5%** in Data Mining Kaggle contest at BITS Pilani.
- Received the **Team Excellence Award** for outstanding performance as an intern at Couture.ai.